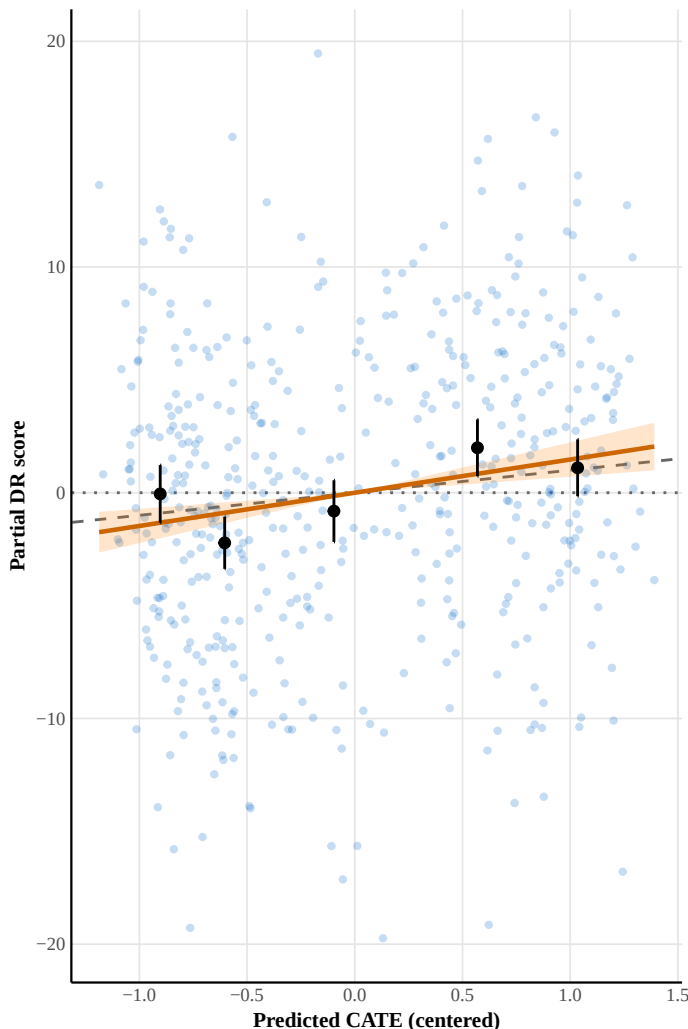


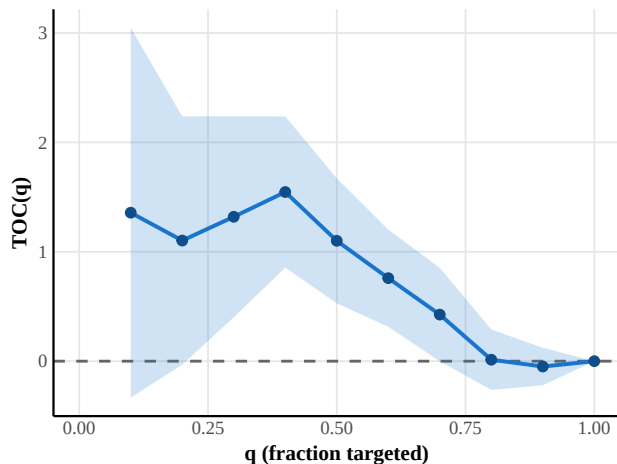
BLP calibration test (Chernozhukov et al., 2018)

slope = 1.472 (SE 0.387), $p < 0.001$ one-sided; reject H_0 .
near dashed = calibrated; near dotted = no heterogeneity.



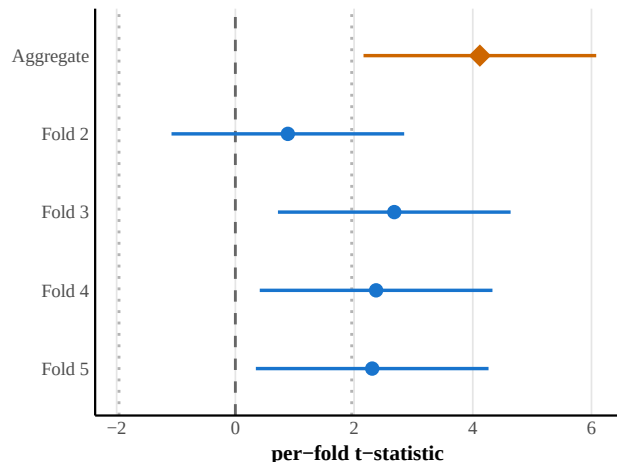
OOB TOC curve (AUTOC)

AUTOC = 0.827 (SE 0.273), $p = 0.002$ (two-sided).
Curve above 0 = top-ranked units gain more than average.



Sequential RATE (Wager, 2024)

agg. $t = 4.119$, $K = 5$; $p < 0.001$ two-sided; reject H_0 .
| t_k | > 1.96 (dotted) = fold individually significant.



Note: LEFT panel: Best Linear Predictor calibration test (Chernozhukov et al. 2018). Regresses doubly-robust (AIPW) treatment-effect scores on the forest's out-of-bag predicted Conditional Average Treatment Effect (CATE), centered at its mean. Slope near 1 (dashed reference) = well-calibrated heterogeneity; slope near 0 (dotted) = no detectable heterogeneity. Black points: within-bin mean partial residuals with 95% intervals. TOP-RIGHT: Out-of-bag Targeting Operator Characteristic (TOC) curve (Yadlowsky et al. 2024). $TOC(q)$ = Average Treatment Effect among top- q units ranked by predicted CATE, minus the overall Average Treatment Effect; curve above zero at small q = top-ranked units gain more than average. AUTOC / QINI summarize the curve as a scalar. Heuristic: out-of-bag-based, anti-conservative. BOTTOM-RIGHT: Sequential Rank-Average Treatment Effect (RATE) test (Wager 2024). Per-fold t_k = $RATE_k / \text{standard error}_k$ from CATE forests trained on prior folds; blue segments = 95% Wald interval ($t_k \pm 1.96$). Orange diamond = aggregate $t = \text{sum}(t_k) / \sqrt{K-1}$, standard normal under the null.